202606UECM3473OE1b

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Review of preview

Started on	Monday, 29 June 2026, 11:25 AM
Completed on	Monday, 29 June 2026, 11:25 AM
Time taken	9 secs
Grade	**0** out of a maximum of 10 (**0**%)

1

Marks: 1

The random variable X has the density function with parameter β given by

$$f(x;\beta) = (1/\beta^2)x^1e^{-(1/2)(x/\beta)^2},\ x > 0,\ \beta>0.$$

You are given the following observationof X:

4.3, 1.6, 3.3, 6.5, 4.8.

Determine the method of moments estimate of β. [Note: Γ(1+1/2) = 0.8862.] ______

Answer:

Make comment or override grade

Incorrect

Correct answer: 3.271363

Marks for this submission: 0/1.

2
Marks: 1

You are given the following data for claim sizes:

Claim size	Number of claims
Under 1100	14
[1100, 2200)	8
2200 and up	4

The data are fit to an exponential distribution using maximum likelihood. Determine the fitted mean. ______

Answer:

Make comment or override grade

Incorrect
Correct answer: 1271.68
Marks for this submission: 0/1.

3
Marks: 1

An auto liability coverage has a claims limit of 130. Claim sizes observed are

21, 52, 58, 83, 130

where the claim at 130 was for exactly 130. In addition, there are 5 claims above the limit. The data are fitted to an exponential distribution. Determine the MLE of θ. ______

Answer:

Make comment or override grade

Incorrect
Correct answer: 198.8
Marks for this submission: 0/1.

4
Marks: 1

Annual claim counts follow a geometric distriutio with mean β.

- 77 policyholders submitted 0 claims.
- 17 policyholders submitted 1 claim.
- 2 policyholders submitted 2 claims.
- For two policyholders, it is know that they submitted either 1 claim or 2 claims, but the exact number of claims is not available.
- No policyholder submitted more that 2 claims.

Estimate β using maximum likelihood. ______

Answer:

Make comment or override grade

Incorrect
Correct answer: 0.238
Marks for this submission: 0/1.

5
Marks: 1

For an insurance coverage, your department only handles claims below 10,000. Among these claims, you observe 6 claims for 800, 1000, 2000, 2500, 4000, and 5000. In addition, there are 4 claims for amounts below 500, who exact amounts are unknown. You fit these claims to an inverse exponential distribution using maximum likelihood. Determine the resulting estimate for the 65^{th} percentile of claim size for all claims. ______

Answer:

Make comment or override grade

Incorrect
Correct answer: 1161.37
Marks for this submission: 0/1.

6
Marks: 1

8 losses have been recorded in thousands of dollars and are grouped as follows:

Interval	[0, 1)	[1, 8)
Number of Losses	3	5

There is no record of the number of losses at or above 8,000. The random variable X underlying the losses, in thousands, has the density function

$$f(x) = \lambda e^{-\lambda x}, x > 0, \lambda > 0.$$

Determine $L(\lambda)$ and MLE of λ. ______

Answer:

Make comment or override grade

Incorrect
Correct answer: 0.4543
Marks for this submission: 0/1.

7
Marks: 1

For an insurance policy, you are given:

- Ground-up losses follow a Weibull distribution with parameters τ =4 and θ(unknown).
- Losses under 790 are not reported to the insurer.
- For each loss over 790, there is a deductible of 790 and a policy limit of 3100.

- A random sample of six claim payments for this policy is: 295 660 885 1090 2310+ 2310+

where + indicates that the original loss exceeds 3100. Determine the 70th percentile of the ground-up distribution. ______

Answer:

Make comment or override grade

Incorrect
Correct answer: 2819.05
Marks for this submission: 0/1.

8

Marks: 1

You are given:

- Fifty claims have been observed from a lognormal distribution with unknown parameters μ and σ.
- The maximum likelihood estimates are $\hat{\mu} = 7.2$ and $\hat{\sigma} = 1.9$.
- The covariance matrix of $\hat{\mu}$ and $\hat{\sigma}$ is [$c_{11} = 0.0387$, $c_{12} = c_{21} = 0$, $c_{22} = 0.0139$]

Determine the variance of the probability that the next claim will be less than or equal to 5,992 using delta method. ______

Answer:

Make comment or override grade

Incorrect
Correct answer: 0.001119
Marks for this submission: 0/1.

9

Marks: 2

Please click the following link to answer the questions:

https://forms.gle/t8Uw4uHRw8tXRjqr6

Then answer 1 here after submitting the form.
[Note: In order to enter the google form, you must make sure that you login to UTAR acccount. If you see "You need permission", this means that your are not login to UTAR account, switch to UTAR account] _____

Answer:

Make comment or override grade

Incorrect
Correct answer: 1
Marks for this submission: 0/2.